

CS & IT ENGINEERING



'C' Programming

Data Types & I/O Functions

Lecture No.- 03



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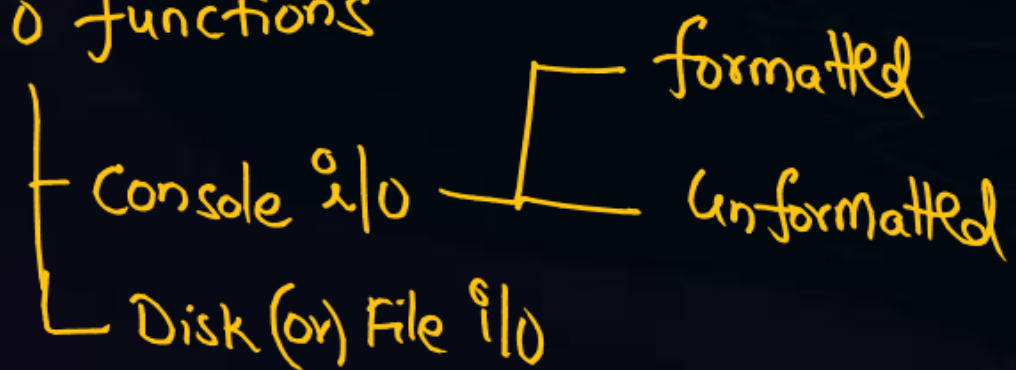
Recap of Previous Lecture



- Primary Data Types in 'C'

- void
- char
- int
- float

- I/O functions



Topics to be Covered



formatted i/o functions

- scanf()
- printf()





Topic : I/O FUNCTIONS IN 'C'



Printf() Function : Pre-defined / System-defined / Library function

- 'f' in Printf() (or) scanf() indicates formatted data

- Syntax:

int	Printf	(char *)
↓	↓	↓
return type	Name of function	input type = character pointer (or) String

⇒ Printf() function process character(s) to the o/p console (monitor) and return number of characters processed as return value.

Ex: `Printf("HELLO");`



returns 5 as return value.

O.S Kernel

Ex:

```
main()
{
    Printf("HELLO");
}
```

return 5

o/p: HELLO

```
main()
{
    int a;
    a = Printf("HAI");
    Printf("%d", a);
}
```

o/p: HAI 3



Topic : I/O FUNCTIONS IN 'C'



Printf()

① `Printf("message");`

Ex: `Printf("HELLO");` o/p: HELLO

`Printf("WELCUM");` o/p: WELCUM

② `Printf("message format Specifier", variable(s));`

→ When data need to be formatted (or) data is retrieved from memory

Ex: `int a = 72;` a ^{2 Bytes} 72

`Printf("a value is %d", a);`

o/p: a value is 72

`Printf("%d is the value of a", a)`

o/p: 72 is the value of a

`Printf("%d", a);`

o/p: 72

%d decimal Integer

Ex: `int a=7, b=3, c=5;`

`Printf("a=%d, b=%d, c=%d", b, c, a);`

o/p: a=3, b=5, c=7



Topic : I/O FUNCTIONS IN 'C'



Ex: `main()`
`printf("%d", printf("HELLO"));`

o/p: HELLO 5

Ex: `int a=123, b=45, c=6`

`printf("%d", printf("%d %d %d", printf("a=%d, b=%d, c=%d", a, b, c))));`

Garbage value

o/p:

1 2 3 4 5 6 7 8 9 10 11 12 13 14
a = 123, b = 45, c = 6 14 Garbage values



Topic : I/O FUNCTIONS IN 'C'



Ex:

```
int a = printf("HELLO");
int b = printf("a = %d", a);
int c = printf("%d", b);
printf("c");
```

O/p: HELLO a=53c

Printf Process() Evaluates in LIFO order (or) Right to Left

Ex:

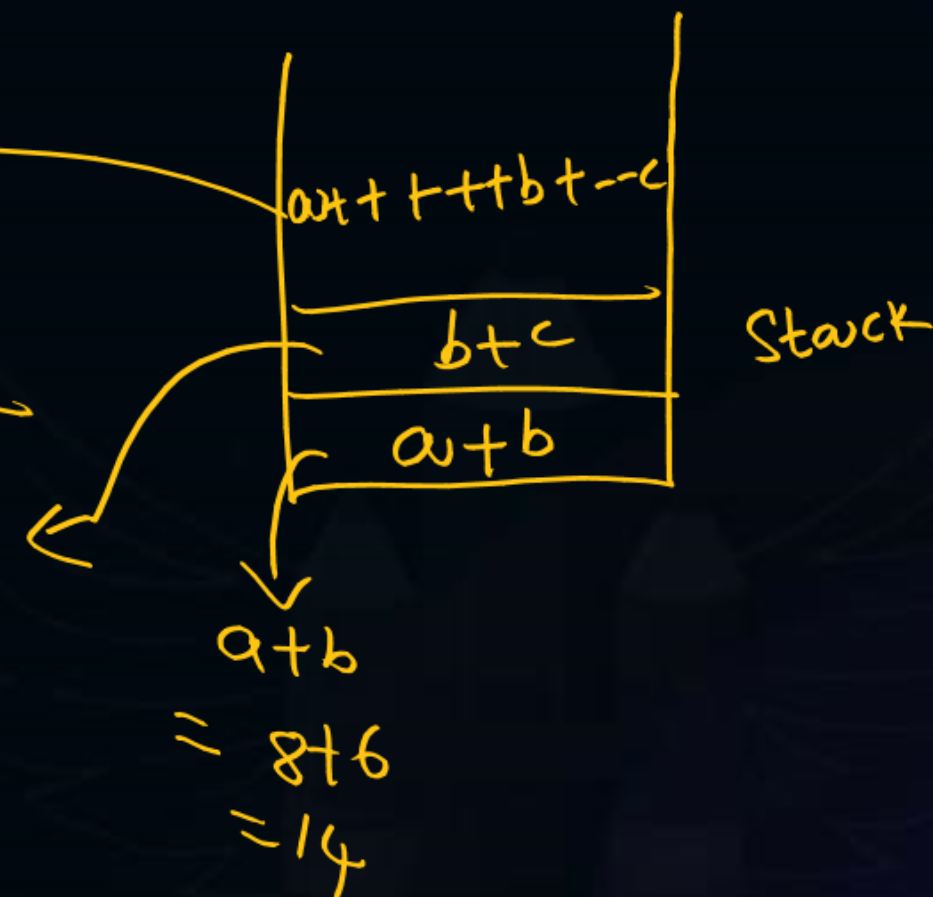
```
int a = 8, b = 6, c = 2, d;
```

```
d = printf("%d %d %d", a+b, b+c, a++ + ++b + --c);
```

$a++ + ++b + --c$
 $= 7 + 6 + 2$

$b+c$
 $= 6 + 2 = 8$

O/p: 14 8 15





Topic : I/O FUNCTIONS IN 'C'

Char c = 'B';

int a = 57, b = 064, c = 0x35

float d = 13.2579;

① printf("%d", a); // 57

② printf("%o", b); // 64

③ printf("%x", c); // 35

④ printf("%c", a); // 'q' (ASCII char of 57)

⑤ printf("%f", ^(float)b); // 52.000000

⑥ printf("%f", ^(float)a); // 57.000000

⑦ printf("%f", ^(float)c); // 53.000000

⑧ printf("%7d", a); // -----57

⑨ printf("%-7d", a); // 57-----

⑩ printf("%f", d); // 13.257900

$$(64)_8 = 6 \times 8^1 + 4 \times 8^0 = 48 + 4 = (52)_{10}$$

$$(35)_{16} = 3 \times 16^1 + 5 \times 16^0 = 48 + 5 = (53)_{10}$$

(destination type) source Explicit Type Casting
int ↔ float

⑪ printf("%.1f", d); // 13.2579

⑫ printf("%.e", d); // 132579e-4

⑬ printf("%d", ^(int)d); // 13

⑭ printf("%.8.2f", d); // 13.25790000

⑮ printf("%.8.3f", d); // 13.2579000

⑯ printf("%c", e); // B

⑰ printf("%d", e); // 66 (ASCII value of 'B')

⑱ printf("%f", ^(float)e); // 66.000000

⑲ printf("%c", ^(char)d); // ASCII char of 13

⑳ printf("%d", c); // 53

$$132579e-4 \Rightarrow 132579 \times 10^{-4} = 13.2579$$



datatype	Format specifier
char	%.c
int, short	%.d (or) %i
unsigned int	%.u
long int	%.ld
Octal integer	%.o
Hexa Decimal Int	%.x (or) %X
float	%.f (or) %e
double	%.f (or) %lf
long long int	%.Ld
string	%.s



Topic : I/O FUNCTIONS IN 'C'



Escape Sequence (special characters)

\n	New line
\t	tab space
\b	Backspace
\r	Carriage return
\0	NULL character

Ex: `printf("HELLO\nAll\tStudents\b");`

Counted as single character.

o/p: HELLO

All ---- Student\$



2 mins Summary



Printf() function

— Explored in detail

To be contd... 



THANK - YOU